

“Luminova”

Circular Light Bulb



01 Why LED Bulbs Need to Change

The scale of the problem

- ~**2 billion** light bulbs sold annually in Europe
- >**5 million** bulbs reach end-of-life every day
- Rapid growth of LEDs → **massive future waste stream**

Linear lifecycle

Production → Use → Disposal

- Complex multi-material product (plastics, PCB, metals)
- **Difficult disassembly → low recycling efficiency**
- Entire bulb discarded even if **only electronics fail**



<https://www.erecycling.ch/fr/ueber-sens/medien-und-publikationen/themendossiers/leuchtmittel.html>



Project objective

Transition from a disposable product to a modular circular system

Power : 8W

Lumens : 800 lm

Mass : 30g



@signify.com

« Luminova » Circular Product Definition

Key Partnerships



Manufacturing & component suppliers

- AT&S

Key Activities



The core work at LumiNova sits at two ends of the product life: **designing it right and getting it back.**

Value Propositions



Modular architecture

Every bulb is designed from the ground up as a system of independent modules: LED board,

Customer Relationships



Adoption depends on trust : users need to believe they can repair this themselves.

Customer Segments



Three very different groups benefit from LumiNova, each in their own way.

Primary target

03



How We Build It

European specialists only : AT&S, Ams OSRAM, Novelis CH, 3B Fibreglass. LumiNova manages assembly, PCB sandwich & optical envelope in-house. University partnerships push LED innovation towards 2050.

01



What We Offer

Fully modular : LED board, driver PCB, housing each replaceable.
Built-in diagnostics, snap-fit self-repair in 5 min.
Recycled Al housing, biodegradable PCB resin without compromising the performance.

04



The Loop

SENSeRecycling handles bulb recovery, LumiNova CH manages PCB & aluminium in-house. Reverse logistics costs shared with partners who benefit from recovery → the circular model pays for itself.

02



How We Make Money

B2B modular sales at higher margin, recurring module replacement, material valorisation across three recovery streams : PCB, Al & components. Modularity drives volume.

Cost Structure

R&D and modular design costs

PCB design equipment and tooling are among the **largest upfront investments**, largely co-funded with industrial partners.

Reverse logistics and recycling costs

Costs shared with partners who directly benefit from material recovery.

Raw materials and bought-in component

Significant share of unit cost, partially offset by reduced price volatility on the recycled fraction.

Support infrastructure

Tutorials and QR system is ongoing cost.

Built-in fault detection

individual replacement modules.

Reverse logistics

Modules are collected back via drop-off points and a take-back programme, feeding directly into recycling loop.

Revenue Streams

Product attractiveness

Productivity drives higher sales volume as customers choose LumiNova over other alternatives.

Modular sale

Manufacturers buy only the component, avoiding full assembly, so higher margin per transaction.

conservation organisations benefit from reduced landfill volume.

EU legislation: The EU Right to Repair directive makes LumiNova's model a compliance advantage as rules tighten through 2030.

Material valorisation

Recovered PCB, aluminium and components resold to recyclers across three separate recovery streams.

EU funding (possibilities)

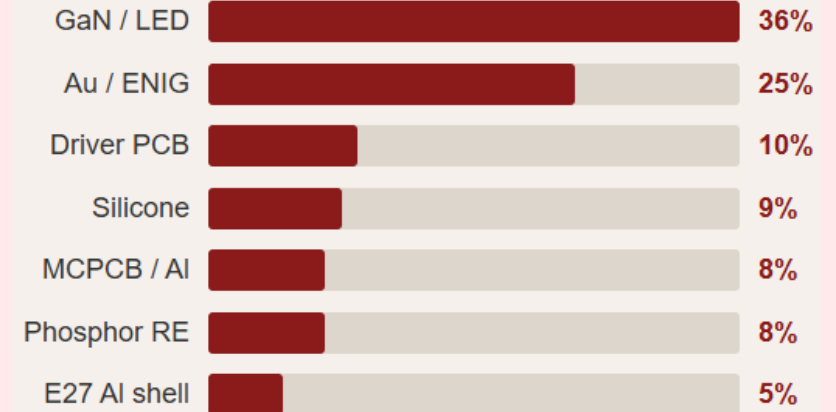
Alignment with the Right to Repair directive opens access to grants, subsidies and public procurement advantages.

Material Selection & Carbon Footprint

A | Scope 3 Upstream — Material CO₂ Hotspots

Component / Material	kgCO ₂ e/kg	Main Scope 3 driver
Driver PCB FR4 + epoxy, Cu, Si IC	13–18	Epoxy resin (fossil), Cu foil electrolysis, Al cap etching
MCPCB (LED board) Al-core + Cu	10–15	Primary Al smelting (Hall-Héroult), Cu electrolysis
LED surface Gold / ENIG finish	~40*	Au mining, cyanide refining, electrolytic deposition
E27 shell Primary aluminium	8.5	Bauxite → Bayer → Hall-Héroult (energy-intensive)
LED die / GaN Gallium nitride	35–80	MOCVD high-temp epitaxy, by-product Zn/Al mining
Encapsulant / lens Silicone resin	6–22	Polysiloxane synthesis, LED lens curing
Phosphor Yttrium oxide (RE)	13	Rare-earth mining (China), solvent extraction

Scope 3 Upstream Share



B | Best Low-Carbon Alternatives (Circular Design)

Recycled Al

Shell + MCPCB

Before: 8.5
After: 2.6 kgCO₂e/kg

–69%

Bio-degradable resin

PCB resin (FR4)

Before: 15
After: 5.0 kgCO₂e/kg

–67%

Au → Au + Ag+ Cu

Au PCB finish

Before: ~40
After: ~ 16 kgCO₂e/kg

–40%

Longer lifetime

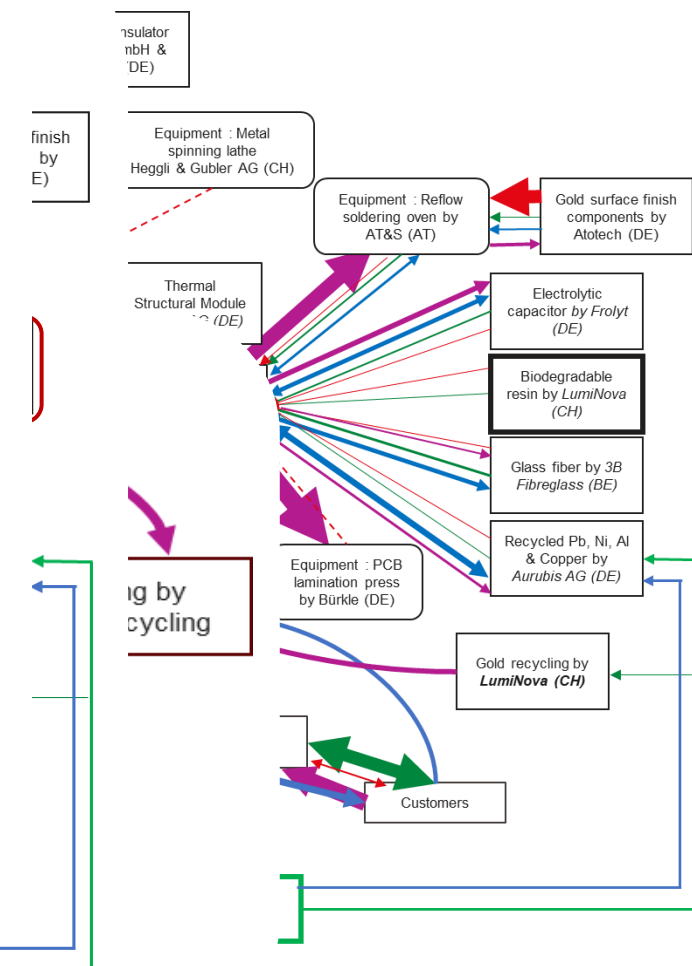
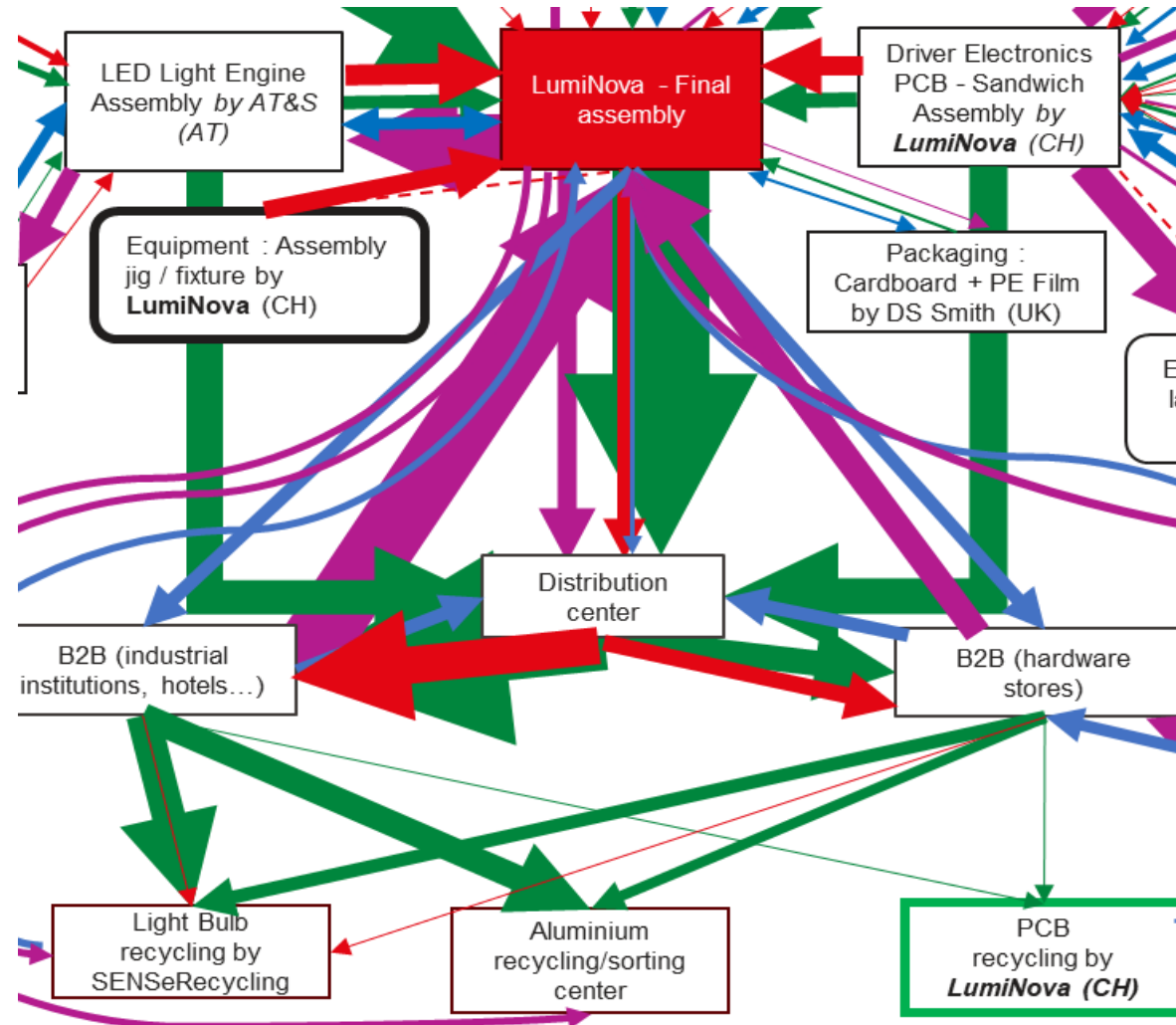
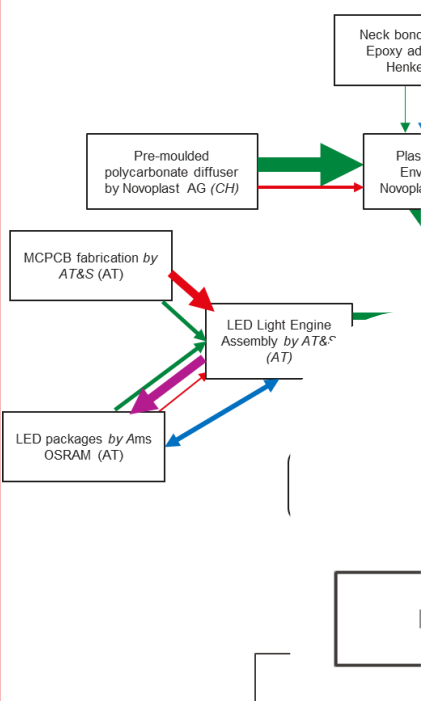
15k → 60k hours

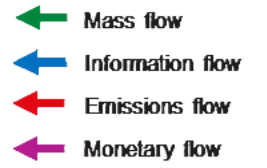
Before: 0.327/b
After: 0.196 kgCO₂e/bulb

–40%

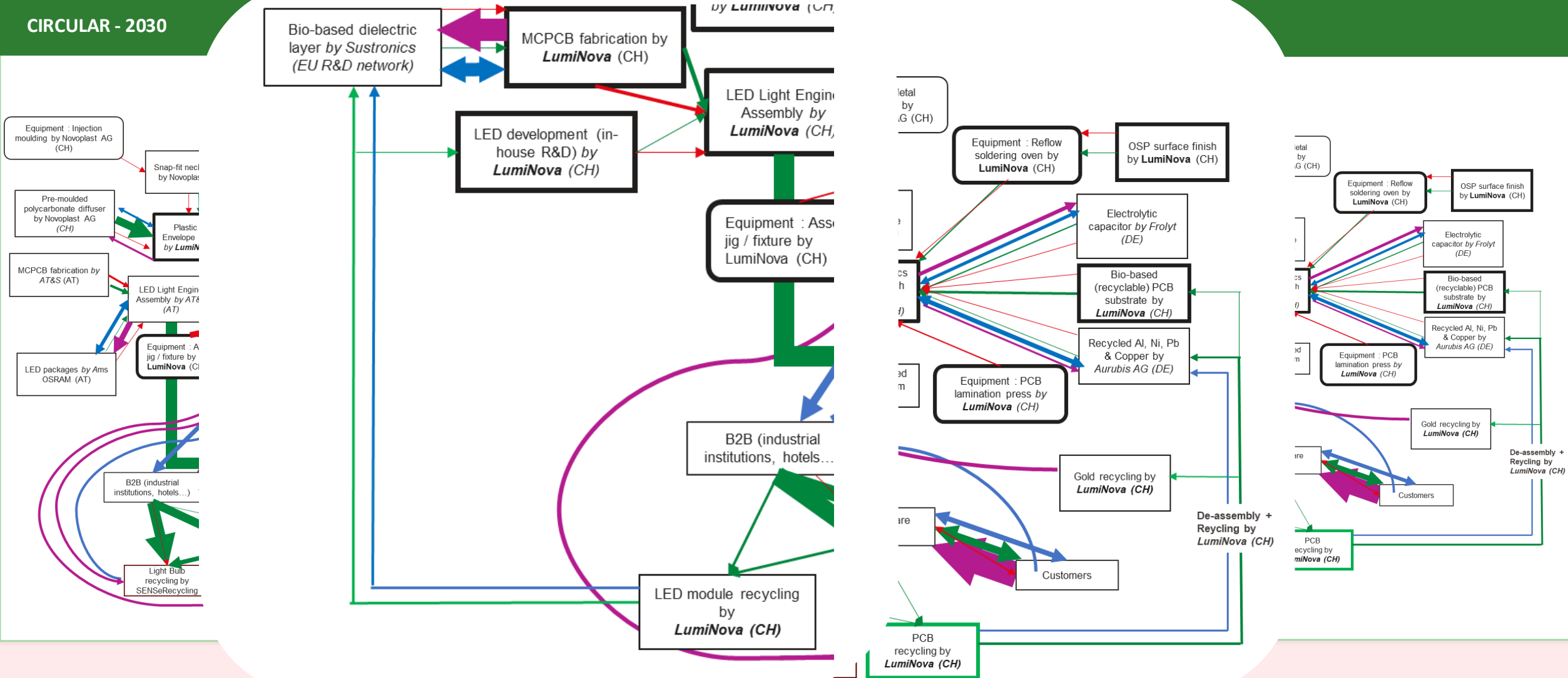
- Mass flow
- Information flow
- Emissions flow
- Monetary flow

LINEAR - Today

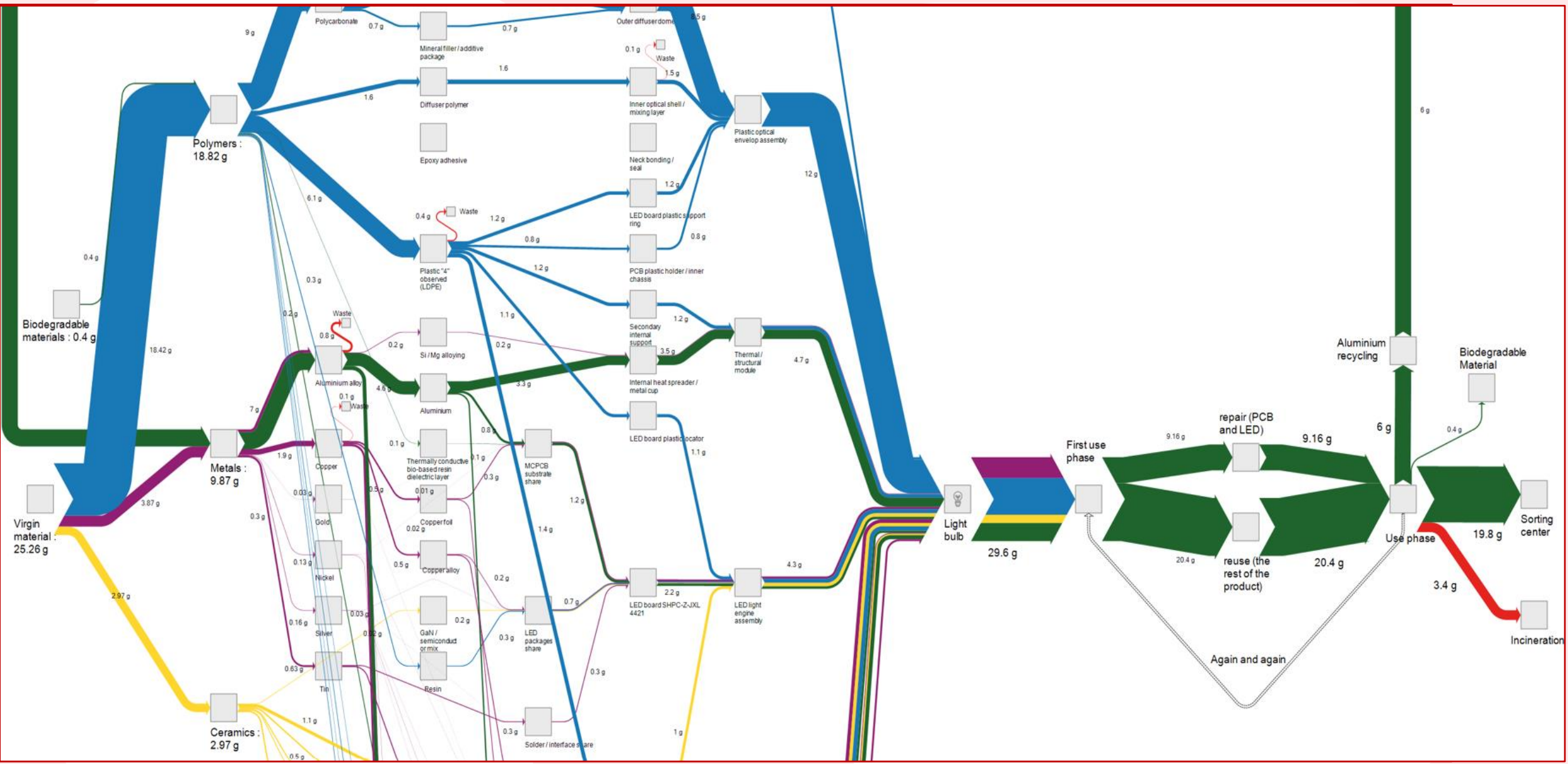




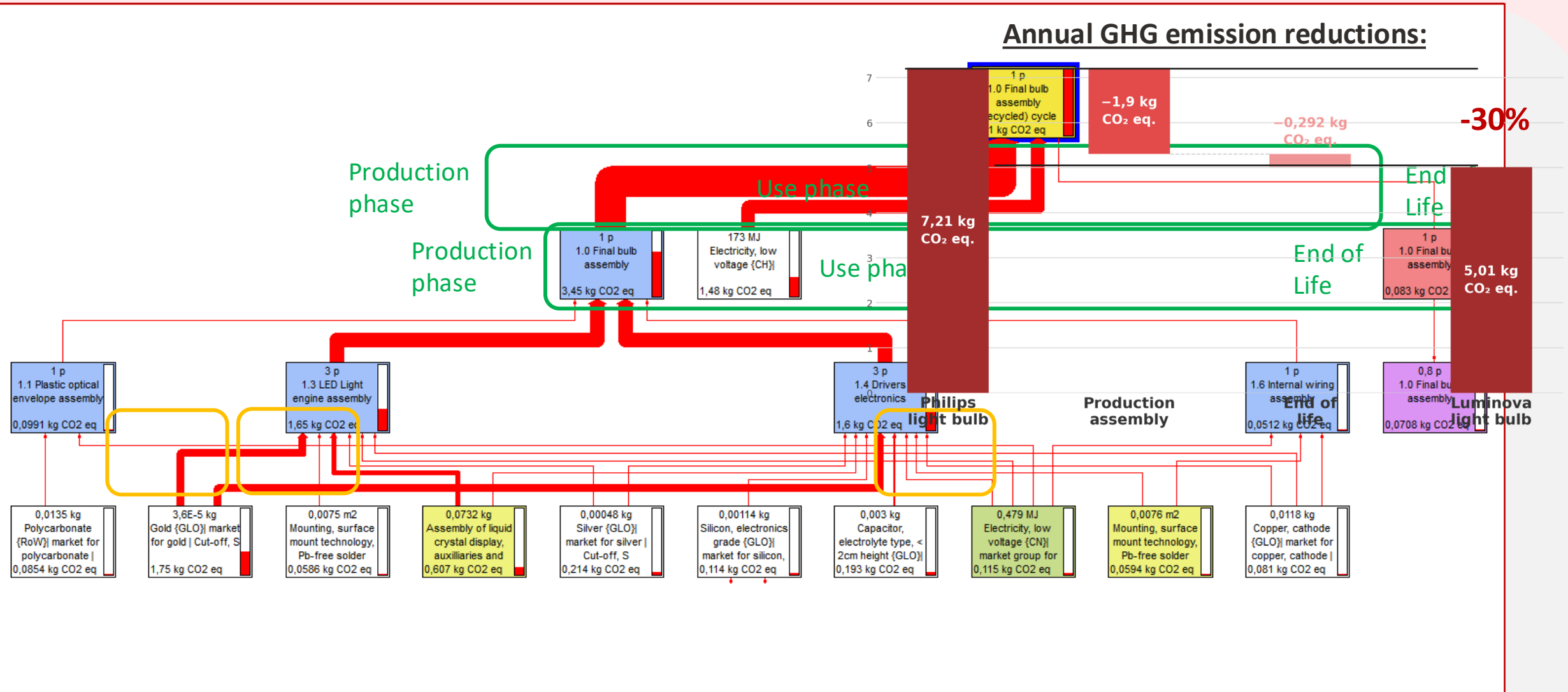
CIRCULAR - 2030



Circular Material Flow Analysis

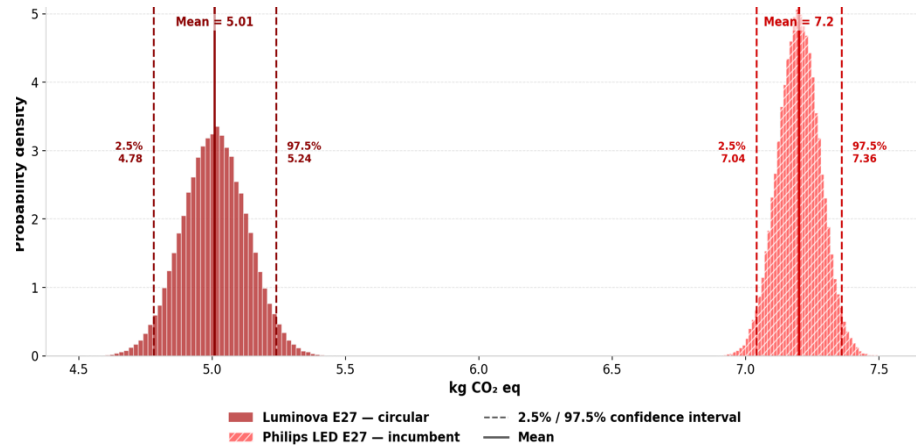


Comparative LCA performed in SimaPro ecoinvent models, based on a **60,000-hour lifetime scenario**.

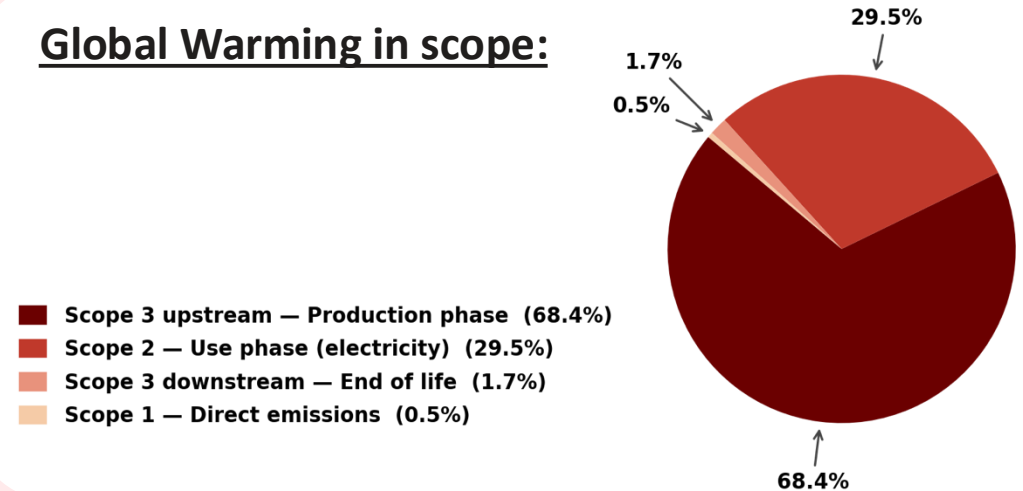


Life Cycle Assessment

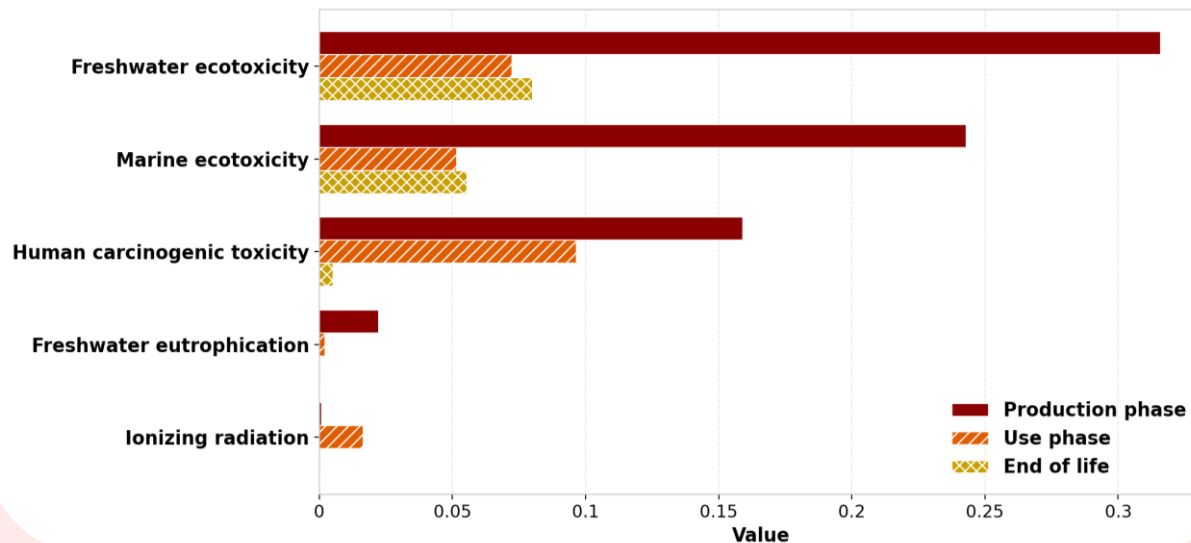
Uncertainty analysis – Global warming:



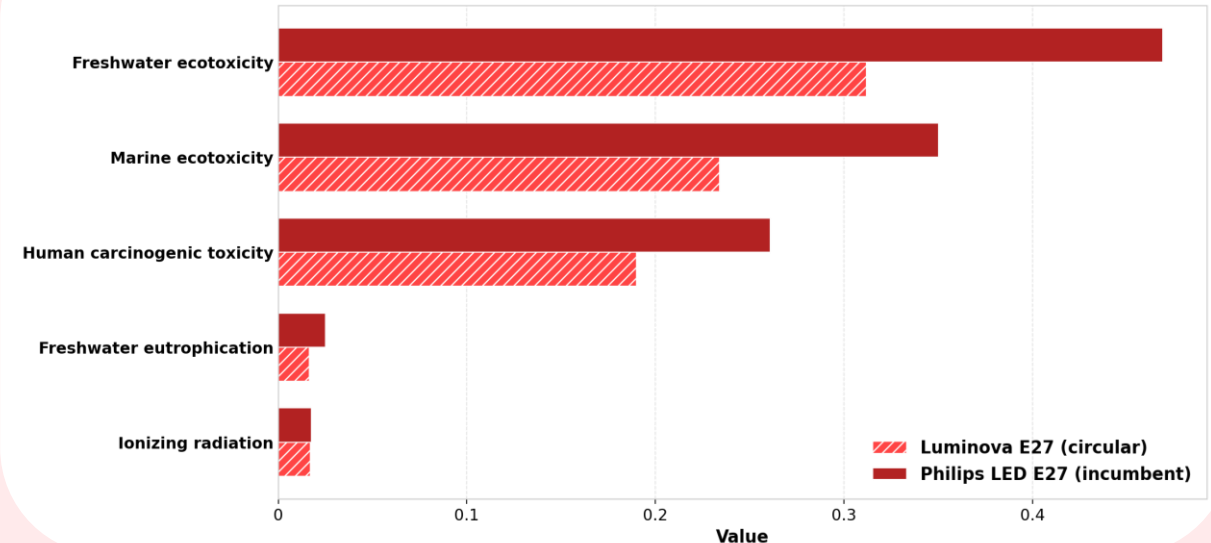
Global Warming in scope:



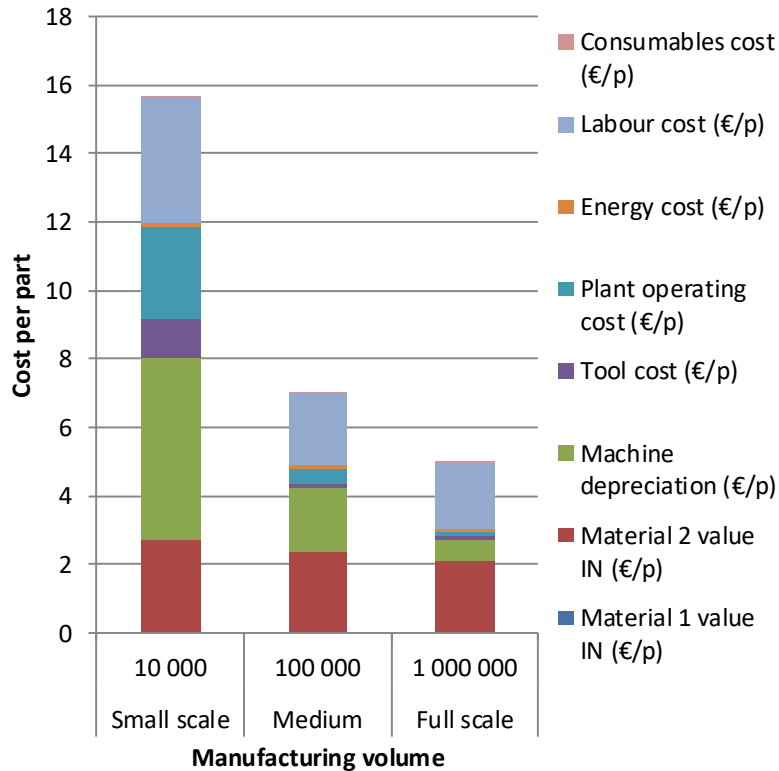
Life Cycle Impact Assessment Philips LED A60 E27:



Life Cycle Impact Assessment Philips LED vs Luminova :

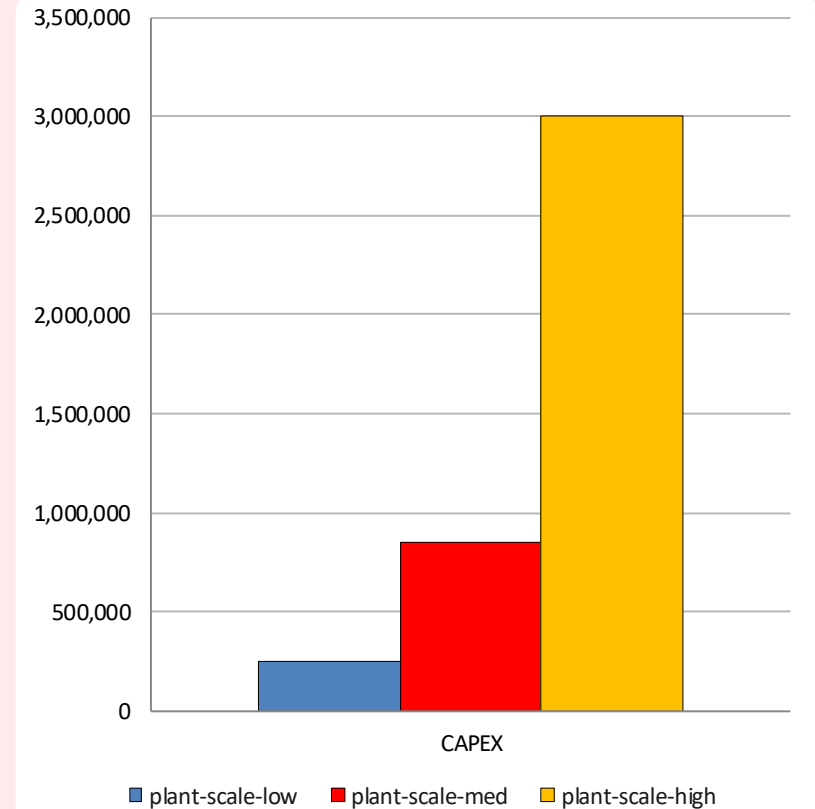


Cost Model & Capex Analysis



Compute with "Cost modeling tool" excel

- Very high sensitivity to scale
- Decreasing price in materials for high scale
- Labour cost high due to more operator at full scale
- Goal : Medium-scale
→ **Selling price of 9,99 €**
9,35 CHF
(Margin of 25%)



Compute with "Cost modeling tool" excel



Replacement LED or PCB at **1,99 €**
1,85 CHF

Customer investment
return **after the 1st repair**

Initiatives, their barriers and how to overcome them

Design for Disassembly (2030)

Main barrier: customer compliance

Strategy: disassembly/repair instructions, failure detection



Carlo Giardinetti

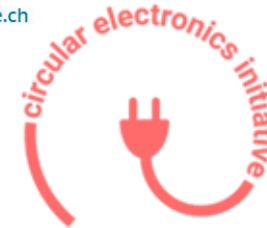
Director, Product Innovation & Circularity

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Milani

Volutio.



Shape the future
with confidence

Anis Nassar

Lead, Circular Economy Innovation and Business Engagement, World Economic Forum



Your Sustainability Consulting Partners

Biodegradable PCB resin (2030)

Main barriers: material performance, manufacturing compatibility, limited long-term data

Strategy: in-house fabrication, prototypes and reliability testing, collaborate with material suppliers

ADVANCED SUSTAINABLE SYSTEMS

Review | [Open Access](#) | [CC BY](#)

Biodegradable Substrates for Rigid and Flexible Circuit Boards:
A Review

Karel Dušek, Daniel Koc, Petr Veselý, Denis Froš, Attila Géczy

First published: 16 October 2024 | <https://doi.org/10.1002/adsu.202400518> | [VIEW METRICS](#)



JIVA

Initiatives, their barriers and how to overcome them

Recycle + reduce gold (2030), eliminate gold (2050)

Main barriers: cost, PCB performance and shelf-life

Strategy: evaluate + implement recycling techniques, redesign + test OSP substitutes



Waste Management
Volume 216, 20 April 2026, 115467



Research Paper

Economic feasibility and profitability analysis of scCO₂-based gold recovery process from waste printed circuit boards

Xiangmei Yang ^a, Min An ^b



David M. Spitz

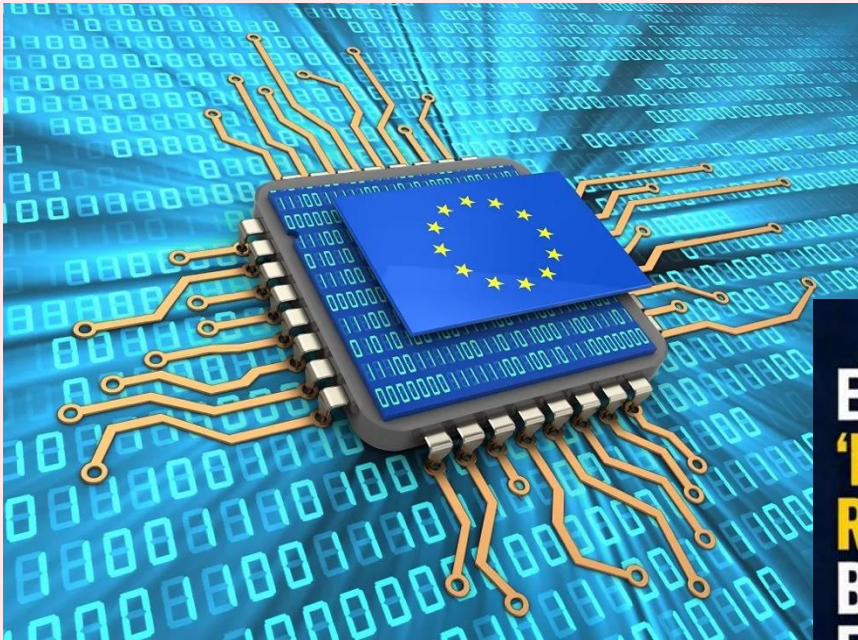
30 years experience in
PCBA Manufacturing
Technical leadership in
DFM Engineering

Local production of PCBs (2030), LED chips (2050)

Main barriers: cost, supply/capacity limitations, non-trivial scaling

Strategy: DFM, supplier qualification, dual-sourcing, phased capacity building





<https://www.eetimes.com/eu-council-adopts-position-on-chips-act/>

7.21 kg → 5.01 kg CO₂ eq.
15 k → 60 k hours
24 % → 70 % circularity

Our redesign is a response to the future of manufacturing



<https://techstory.in/eu-wants-replaceable-batteries-but-iphones-might-not-need-them/>

EPFL

able materials

Annexes

■ École
polytechnique
fédérale
de Lausanne



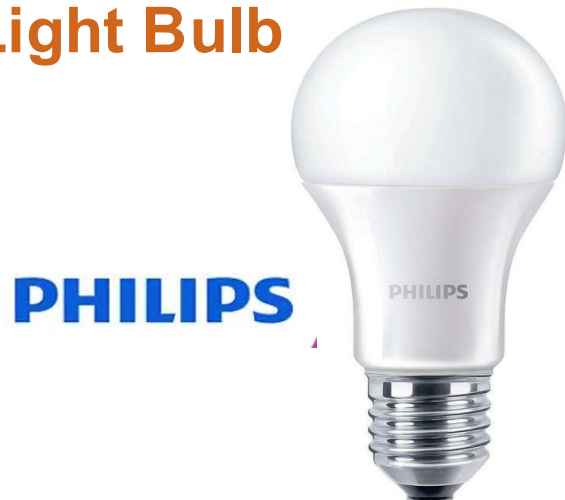
PRESENTED BY
Lea Goutal; Elias Baiada; Thomas
Kuntze; Rémy Grillet-Aubert;
Mathélyo Favre-Régullon

Objectives: From Conventional LED Bulbs to a Modular, Circular Design

In Europe, nearly **2 billion light bulbs** are sold yearly; equivalent to more than **5 million bulbs** reaching end-of-life every day

1.

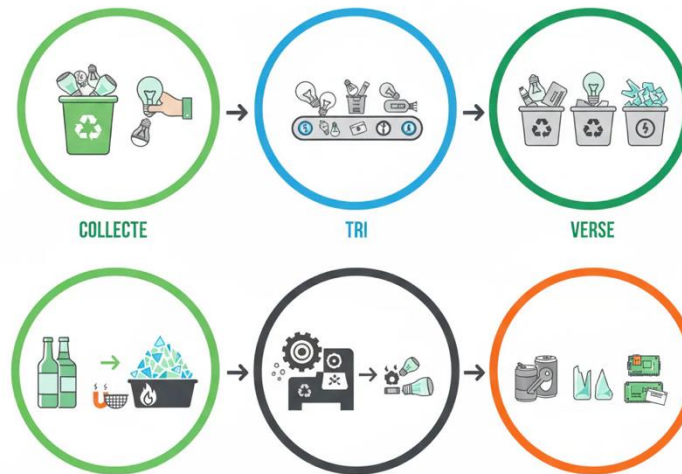
Environmental Impact of
**Conventional Philips
Light Bulb**



<https://www.lighting.philips.fr/consumer/choose-a-bulb>

2.

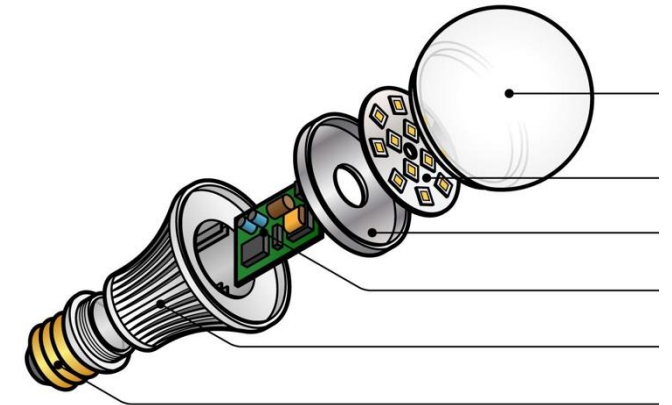
Analysis of Current
Circularity Bottlenecks



<https://www.silamp.fr/pages/recyclage-des-ampoules-led-ou-en-est-on>

3.

Propose a **Modular
Redesign**



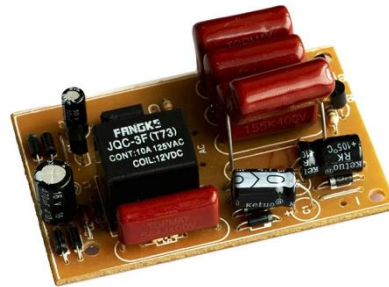
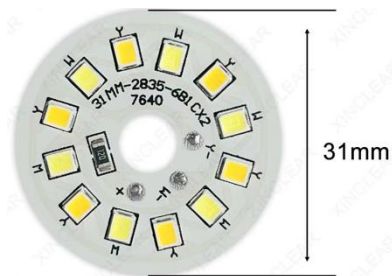
<https://www.iea.org/commentaries/the-next-wave-of-led-lighting-smarter-circular-and-more-efficient>

3. « LumiNova » – Circular Product Redesign

Design, Functionality & Materials Innovations

Design & Functionality Improvements:

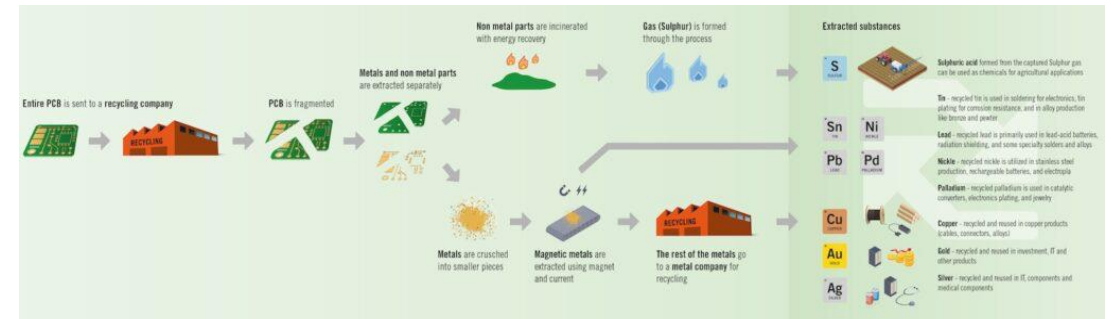
- Fully modular architecture
- Easy disassembly of all components (PCB, base, housing)
- **Replace-only strategy:** defective parts can be individually replaced (*LED board, driver PCB, etc.*)



- **Integrated failure detection system**
→ identifies whether the LED module or driver is faulty
- **Reduced overconsumption** through extended product lifespan

Material Optimization Strategy

- Reduced plastic content
- **Use of recycled metals**
→ recycled aluminium & recycled copper
- **Gold minimization strategy**
→ reduced, recycled, or substitution with alternative materials
- **Eco-designed PCB**
→ improved recyclability
→ biodegradable resin



4. Circular CBM & Safe Product Redesign

4.1 Circular CBM

Modular LED bulb :

1. Replace only the LED plate/module :
Less e-waste
2. Universal socket compatibility



Values Proposition

- Only the defective module is replaced, not the whole bulb
- Compatible with all standard sockets (E27, E14, GU10)
- Longer lifespan, less electronic waste



Revenue Streams

- Universal base sold to manufacturers (B2B)
- Replacement modules sold to consumers (recurring)



Key Partners

- Lighting manufacturers, recycling companies, retailers



Costs

- R&D, manufacturing, reverse logistics & recycling

4.2 Safe & Circular Product Redesign Workshop

CHEMICALS TO DESIGN OUT

Toxic Epoxy resin

PCB / adhesive - 25068-38-6
→ Replace with biodegradable resin

Toxic Ethylene glycol

Capacitor - 107-21-1
→ Replace only when faulty via smart diagnostic

Critical Gallium nitride

LED chip - 25617-97-4
→ Minimise via replaceable LED module

Finite Rare earth mix

Phosphor coating - Ce 7440-45-1 & Te 7440-27-9
→ Reduce via alternative material strategy

3 REDESIGN CONCEPTS

Universal modular system

Replaceable LED, driver and shell — reduces GaN and rare earth consumption over lifetime.

GaN + rare earths

Glue-free service bulb

Screws replace adhesive — eliminates epoxy resin, enables full PCB disassembly.

Epoxy resin

Smart diagnostic light

Indicator identifies LED vs PCB fault — replaces capacitors only when strictly needed.

Ethylene glycol



Safe & Circular Product Redesign Workshop

Designed for : LumiNova - Circular LED
Bulb

Designed by :
Rémy Grillet-Aubert

Date : 10/04/2026

Version : II

DEFINE THE CHEMICALS OF CONCERN (15 MINS)

PICK & SCREEN TWO CHEMICALS THAT ARE IN THE PRODUCT

This is a summary of all the chemicals that can be found in a product like this, and not a complete list.
Screen the different chemicals using [MaterialWise.org](https://www.materialwise.org).

WHICH CHEMICALS SHOULD BE DESIGNED OUT?

Highlight the chemicals of concern that should be designed out in the next phase.

EXPLORE (5 MINS)

CHOOSE A COMMERCIAL APPLICATION

Philips LED A60 E27 bulb (SKU 8720169191655)

WHAT ARE THE FUNCTIONAL NEEDS AND REQUIREMENTS OF A LIGHT BULB?

INTERVIEW A USER

Core function of the light bulb:

- Provide efficient, safe, and long-lasting illumination

User needs:

- Reliable lighting
- Low energy consumption
- Long lifetime
- Easy replacement
- Affordable cost

☐

PCB / ADHESIVE

| 25068-38-6

TOXIC: Y / ☒ N

EPOXY RESIN

☐

ELECTROLYTE (CAPACITOR)

|

107-21-1

TOXIC: Y / ☒ N

ETHYLENE GLYCOL

☐

LED SEMICONDUCTOR
GALLIUM NITRIDE

|

7440-50-8

TOXIC: Y / N ☐

☐

PHOSPHOR COATING
RARE EARTH MIX

|

7440-45-1

TOXIC: Y / N ☐

☐

GOLD (PCB PLATING)
ELECTROLESS NICKEL IMMERSION GOLD (ENIG)

| 7440-57-5

TOXIC: Y / N ☐



Safe & Circular Product Redesign Workshop

REDESIGN (30 MINS)

DEVELOP 3 CONCEPTS FOR IMPROVEMENTS

Are there different or better ways to meet the needs while designing out the substances of concern? Brainstorm to come up with new ideas using the Material Health Strategy Cards

Describe and illustrate 3 concepts for improvements by using the Material Health Strategy Cards. Try to be visual and give it a memorable name.

CONCEPT IDEA

CONCEPT 1: Universal Modular Lighting System

A modular light bulb with a universal base compatible with different lighting systems beyond standard E27 sockets. The bulb is divided into replaceable modules: LED, driver and outer shell. By making the LED module replaceable, we directly reduce gallium nitride and rare earth consumption over the product lifetime, since fewer LED chips need to be manufactured. This also reduces indirect waste by avoiding full bulb replacement when changing lighting systems.

CONCEPT 2: Glue-Free Service Bulb

A glue-free bulb designed with mechanical connections: clips and screws instead of adhesive. This directly eliminates epoxy resin, a toxic and non-biodegradable substance that prevents PCB separation and recycling. The bulb can be fully disassembled, and components are collected, repaired or recycled through a take-back system. Removing epoxy resin also makes it possible to recover copper and aluminium from the PCB at end of life.

CONCEPT 3: Smart Diagnostic Light

A bulb with an integrated failure detection system that identifies whether the LED plate or the PCB driver needs replacing. Since the PCB contains ethylene glycol in its capacitors, a toxic electrolyte, the goal is to replace it only when strictly necessary. A simple current bridge triggers an indicator light when the PCB is faulty, avoiding unnecessary replacement and reducing toxic waste over time.

CONCEPT 4: Gold-Free PCB Surface Finish

A bulb PCB designed with OSP (Organic Solderability Preservative) surface finish instead of the conventional ENIG (Electroless Nickel Immersion Gold) plating. Gold, despite its negligible mass (0.000006 kg per bulb), carries an extremely high carbon footprint of 3767 kgCO₂e/kg, making it one of the most environmentally damaging materials per gram in the entire product. By switching to OSP, developed in-house by LumiNova, the process eliminates gold and nickel plating entirely. OSP is non-toxic, biodegradable and fully compatible with the bio-based PCB substrate. This change also simplifies end-of-life recovery, as the absence of noble metal plating makes copper extraction from recycled PCBs significantly cleaner.

REFLECT (10 MINS)

WHAT NEEDS TO BE INCLUDED IN THE DESIGN PROCESS?

What data would be important to have? What infrastructure is needed? Who would you need to collaborate with?

Designing a Modular Lighting System: A Holistic Approach

Designing a modular lighting system requires precise data at every level. Understanding the lifespan of each component is essential as LEDs fail faster than PCBs, so replacement cycles must be anticipated. Material toxicity data is equally important, particularly knowing where epoxy resin, ethylene glycol and gallium nitride are used in order to prioritise substitutions. Changing from glued to screw-based assembly or switching to biodegradable resin will also directly impact manufacturing processes, tooling and production costs, so this data must be collected early in the design phase.

In terms of infrastructure, new assembly lines adapted for modular manufacturing will be needed, as well as return and collection points for end-of-life modules. Recycling facilities capable of separating PCB materials, made possible once epoxy resin is eliminated, are also critical. A digital tracking system via QR codes would allow users and technicians to monitor component history and identify failures quickly.

Finally, collaboration is at the core of this approach. Material experts are needed to find substitutes for toxic chemicals. LED manufacturers must work on reducing gallium nitride usage per chip. Electronics manufacturers need to adapt the PCB assembly process. Recycling companies ensure recovered materials re-enter production. And retailers together with repair networks manage module collection and replacement at the consumer level.

Key Partnerships



Manufacturing & component suppliers

- AT&S
- Heggli & Gubler AG
- Ams OSRAM
- Frolyt

Material suppliers

- Novelis CH
- Aurubis AG
- 3B Fibreglass.

End-of-life partners

- SENSErecycling
- Aurubis AG

Academia

University partnerships co-develop the eco-PCB and fault-detection algorithms, with potential for LED innovation towards 2050.

Key Activities



The core work at LumiNova sits at two ends of the product life: **designing it right and getting it back.**

R&D centres on the eco-designed PCB and fault-detection system. LumiNova manages final assembly, PCB sandwich assembly and optical envelope assembly in-house, with AT&S on sub-assembly. Used modules are collected via retail drop-off points and returned for PCB & aluminium recovery.

Key Resources



R&D team

Designs and develops the modular bulb system

Modular architecture

Every component is designed to be returned, repaired or recycled back into the system

Collection network

Turns end-of-life modules into a renewable input rather than waste

Value Propositions



Modular architecture

Every bulb is designed from the ground up as a system of independent modules: LED board, driver PCB, and housing. **When one fails, you replace that part only.** **≈80%** of bulbs discarded due to single component failure

Self-repair, no specialist

A snap-fit neck replaces glue, enabling clean disassembly at end of life. A QR code on the bulb walks the user through a 3-step replacement in under 5 minutes.

Recycled & eco materials

The housing uses recycled aluminium from Novelis CH. The PCB uses biodegradable resin and minimises precious metals. Choosing LumiNova is not a compromise, **it performs as well as conventional alternatives.**

Built-in fault detection

A simple diagnostic circuit identifies whether the LED module or the driver PCB has failed, before the user even needs to open the bulb. No guessing, no buying the wrong part. **2 modules** = only two possible failure points to diagnose

Customer Relationships



Adoption depends on trust : users need to believe they can repair this themselves.

Visual disassembly tutorials

Critical adoption lever
A **QR code** on the bulb lets users diagnose faults and order the right module directly. To close the loop, customers get a discount when they return a dead module instead of buying a new bulb.

Channels



Sales channels

LumiNova reaches its customers through hardware stores, specialist lighting shops, direct B2B outreach to hotels, institutions and large companies, and online for individual replacement modules.

Reverse logistics

Used modules are collected back via retail drop-off points and a take-back programme, feeding directly into the recycling loop.

Customer Segments



Three very different groups benefit from LumiNova, each in their own way.

Primary target

Institutions, industrials, homeowners & renters wanting lower long-term costs and less e-waste. Core revenue.

Retailers Hardware stores

Carry both the LumiNova modular bulb and its individual replacement modules

Recyclers

Companies receiving end-of-life modules for PCB & aluminium material recovery.

Other beneficiaries

Waste management and conservation organisations benefit from reduced landfill volume.
EU legislation: The EU Right to Repair directive makes LumiNova's model a compliance advantage as rules tighten through 2030.

Cost Structure



R&D and modular design costs

PCB design equipment and tooling are among the **largest upfront investments**, largely co-funded with industrial partners.

Reverse logistics and recycling costs

Costs shared with partners who directly benefit from material recovery.

Raw materials and bought-in components

Significant share of unit cost, partially offset by reduced price volatility on the recycled fraction.

Support infrastructure

Tutorials and QR system is a low ongoing cost.

Revenue Streams



Product attractiveness

Modularity drives higher sales volume as customers choose LumiNova over throwaway alternatives.

B2B modular sale

Fixture manufacturers buy only the faulty component, avoiding full reassembly so higher margin per transaction

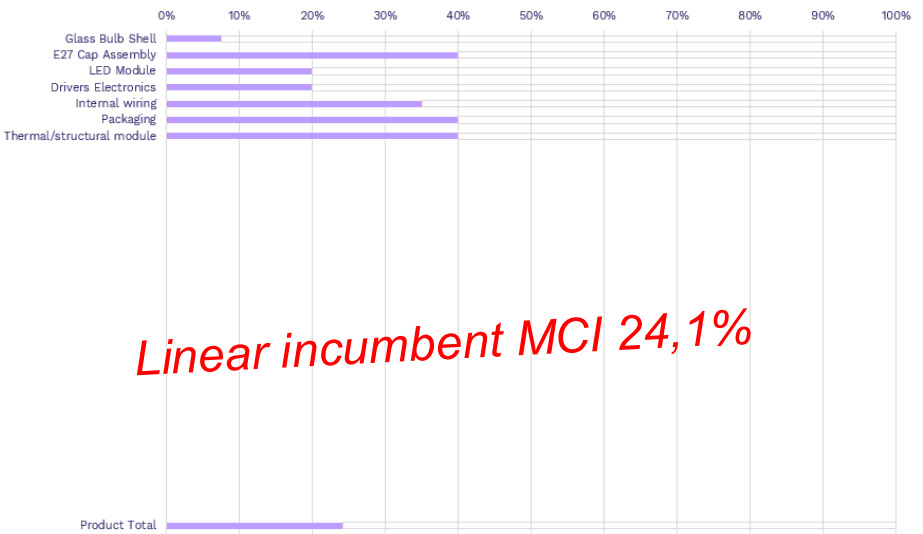
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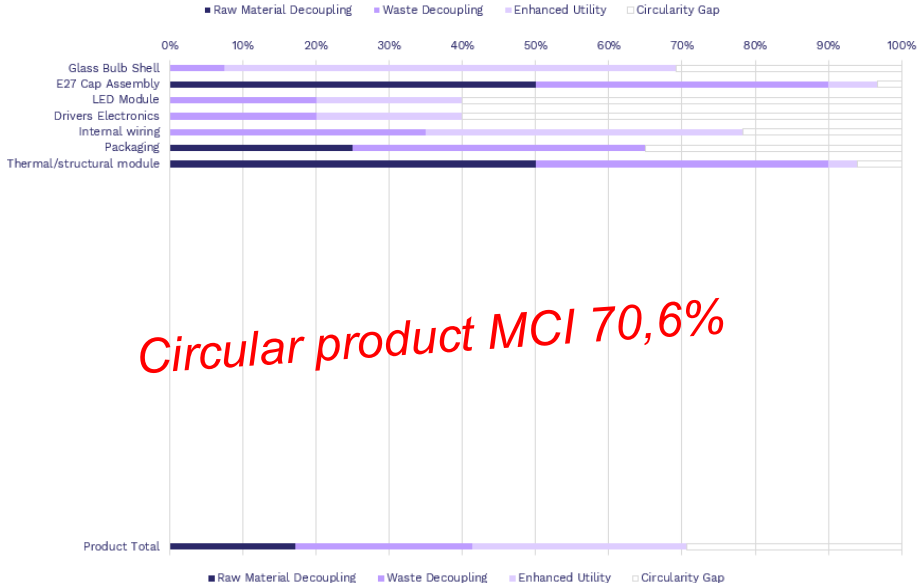
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Alignment with the Right to Repair directive opens access to grants, subsidies and public procurement advantages.

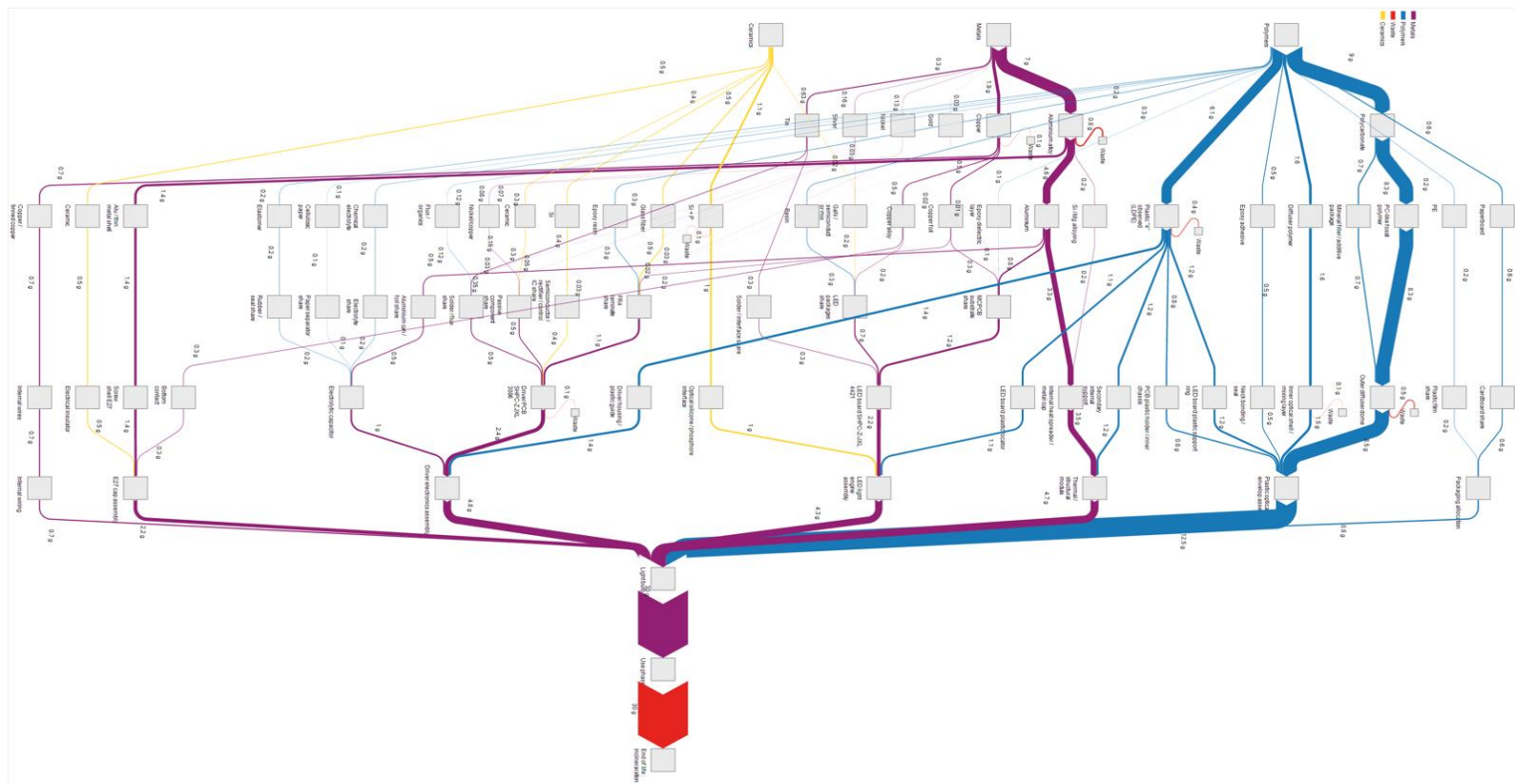
5. Material Circularity Indicator



Linear incumbent MCI 24,1%



Circular product MCI 70,6%



MFA (rotated by 90°) of the incumbent light bulb

EPFL

possible materials

Bill of Material

■ École
polytechnique
fédérale
de Lausanne



PRESENTED BY
Lea Goutal; Elias Baiada; Thomas
Kuntze; Rémy Grillet-Aubert;
Mathélyo Favre-Régullon

Philips LED A60 E27 Bulb

Item No.	Level	Parent Item	Description	Material	Direct mass (kg)	Roll-up mass (kg)	kgCO2e/kg	kgCO2e/part	Roll-up kgCO2e	Manufacturing method (LCA style)	Supplier model	Shipping detail
1.0	0	-	Philips LED A60 E27 bulb, SKU 8720169191655	Mixed	0.0000	0.0300	-	0.0000	0.2190	Final assembly + electrical test + pack-out	Signify China final assembly	Consumer SKU assembled in China; downstream
1.1	1	1.0	Plastic optical envelope assembly	Polymer	0.0000	0.0125	-	0.0000	0.0399	Optical dome molding + bonding + internal plastic supports	APAC plastics supplier → China plant	500 km road supplier → CN plant
1.1.1	2	1.1	Outer diffuser dome	Polycarbonate / diffuser polymer	0.0085	0.0085	3.50	0.0298	0.0298	Injection molding + surface frosting	APAC plastics supplier	500 km road to CN assembly
1.1.1.1	3	1.1.1	Diffuser polymer feed	Polycarbonate family	0.0000	0.0085	-	0.0000	0.0298	Polymer feed preparation	Integrated upstream	Included upstream
1.1.1.1.1	4	1.1.1.1	Base polymer fraction	PC-like fossil polymer	0.0078	0.0078	3.50	0.0273	0.0273	Polymerization	Polymer supplier	Upstream integrated
1.1.1.1.2	4	1.1.1.1	Diffusion / opacity additives	Mineral filler / additive package	0.0007	0.0007	3.50	0.0025	0.0025	Compounding	Additive supplier	Upstream integrated
1.1.2	2	1.1	Inner optical shell / mixing layer	Diffuser polymer	0.0015	0.0015	3.50	0.0053	0.0053	Thin-wall injection molding	APAC plastics supplier	500 km road to CN assembly
1.1.3	2	1.1	Neck bonding / seal	Epoxy adhesive	0.0005	0.0005	2.70	0.0014	0.0014	Adhesive dispensing + curing	APAC chemical supplier	500 km road to CN assembly
1.1.4	2	1.1	LED board plastic support ring	Plastic "4" observed	0.0012	0.0012	2.30	0.0028	0.0028	Injection molding	APAC plastics supplier	500 km road to CN assembly
1.1.5	2	1.1	Driver PCB plastic holder / inner chassis	Plastic "4" observed	0.0008	0.0008	2.30	0.0018	0.0018	Injection molding	APAC plastics supplier	500 km road to CN assembly
1.2	1	1.0	Thermal / structural module	Metal + polymer	0.0000	0.0047	-	0.0000	0.0326	Internal heat spreading + structural support	APAC supplier → China plant	500 km road supplier → CN plant
1.2.1	2	1.2	Internal heat spreader / metal cup	Aluminium alloy	0.0035	0.0035	8.50	0.0298	0.0298	Stamping / deep drawing / forming	APAC metal supplier	500 km road to CN assembly
1.2.1.1	3	1.2.1	Aluminium stock	Aluminium sheet / cup stock	0.0000	0.0035	-	0.0000	0.0298	Sheet preparation	Integrated upstream	Included upstream
1.2.1.1.1	4	1.2.1.1	Primary aluminium share	Aluminium	0.0033	0.0033	8.50	0.0281	0.0281	Smelting + rolling	Aluminium supplier	Upstream integrated
1.2.1.1.2	4	1.2.1.1	Alloying fraction	Si / Mg alloying	0.0002	0.0002	8.50	0.0017	0.0017	Alloying	Alloy supplier	Upstream integrated
1.2.2	2	1.2	Secondary internal support	Plastic "4" observed	0.0012	0.0012	2.30	0.0028	0.0028	Injection molding	APAC plastics supplier	500 km road to CN assembly
1.3	1	1.0	LED light engine assembly	Electronics + polymer	0.0000	0.0043	-	0.0000	0.0403	LED board + optical interface + locator	APAC LED supplier → China plant	800 km road supplier → CN plant
1.3.1	2	1.3	LED board SHPC-Z JXL 4421	MPCB + LEDs	0.0022	0.0022	15.09	0.0332	0.0332	MPCB fabrication + SMT LED mounting	APAC LED board supplier	800 km road to CN assembly
1.3.1.1	3	1.3.1	MPCB substrate share	Aluminium + dielectric + copper + trace finish n	0.0012	0.0012	10.67	0.0128	0.0128	Aluminium-core PCB manufacture	Integrated upstream	Included upstream
1.3.1.1.1	4	1.3.1.1	Aluminium core	Aluminium	0.00078	0.00078	8.50	0.0066	0.0066	Sheet rolling / punching	Metal supplier	Upstream integrated
1.3.1.1.2	4	1.3.1.1	Dielectric layer	Thermally conductive dielectric / epoxy dielect	0.00010	0.00010	10.00	0.0010	0.0010	Coating / lamination	PCB supplier	Upstream integrated
1.3.1.1.3	4	1.3.1.1	Copper circuit layer	Copper foil	0.00029	0.00029	14.00	0.0041	0.0041	Copper lamination + etching	PCB supplier	Upstream integrated
1.3.1.1.4	4	1.3.1.1	Nickel finish trace	Nickel	0.00002	0.00002	20.00	0.0004	0.0004	Surface finishing / plating	PCB supplier	Upstream integrated
1.3.1.1.5	4	1.3.1.1	Gold finish trace	Gold	0.00001	0.00001	40.00	0.0004	0.0004	Surface finishing	PCB supplier	Upstream integrated
1.3.1.2	3	1.3.1	LED packages share	Semiconductor + package metals + silicone/re	0.00070	0.00070	25.86	0.0181	0.0181	LED package assembly	Integrated upstream	Included upstream
1.3.1.2.1	4	1.3.1.2	Semiconductor die share	Gallium high grade / semiconductor proxy	0.00020	0.00020	35.00	0.0070	0.0070	Epitaxy + dicing	Semiconductor supplier	Upstream integrated
1.3.1.2.2	4	1.3.1.2	Leadframe / package metals	Copper	0.00018	0.00018	20.00	0.0036	0.0036	Stamping + plating	Package supplier	Upstream integrated
1.3.1.2.3	4	1.3.1.2	Package resin / lens	Silicone / resin	0.00028	0.00028	21.67	0.0061	0.0061	Dispensing + curing	Package supplier	Upstream integrated
1.3.1.2.4	4	1.3.1.2	Silver interconnect trace	Silver	0.00003	0.00003	25.00	0.0008	0.0008	Wire bonding / metallization	Package supplier	Upstream integrated
1.3.1.2.5	4	1.3.1.2	Palladium trace	Palladium	0.00001	0.00001	45.00	0.0005	0.0005	Metallization / plating	Package supplier	Upstream integrated
1.3.1.3	3	1.3.1	Solder / interface share	Sn-based solder + minor silver	0.00030	0.00030	7.33	0.0022	0.0022	Reflow soldering	Integrated upstream	Included upstream
1.3.1.3.1	4	1.3.1.3	Tin solder fraction	Tin	0.00028	0.00028	6.00	0.0017	0.0017	Solder alloy preparation	Integrated upstream	Included upstream
1.3.1.3.2	4	1.3.1.3	Silver solder fraction	Silver	0.00002	0.00002	25.00	0.0005	0.0005	Solder alloy preparation	Integrated upstream	Included upstream
1.3.2	2	1.3	Optical silicone / phosphor interface	Silicone + yttrium oxide	0.0010	0.0010	7.50	0.0075	0.0075	Coating / dispensing + cure	APAC LED supplier	800 km road to CN assembly
1.3.2.1	3	1.3.2	Optical silicone fraction	Silicone	0.00085	0.00085	6.50	0.0055	0.0055	Dispensing + cure	APAC LED supplier	Included upstream
1.3.2.2	3	1.3.2	Phosphor fraction	Yttrium oxide	0.00015	0.00015	13.33	0.0020	0.0020	Phosphor mixing / coating	Chemical supplier	Included upstream
1.3.3	2	1.3	LED board plastic locator	Plastic "4" observed	0.0011	0.0011	2.30	0.0025	0.0025	Injection molding	APAC plastics supplier	500 km road to CN assembly
1.4	1	1.0	Driver electronics assembly	Electronics	0.0000	0.0048	-	0.0000	0.0786	Driver PCB + capacitor + plastic guide + assembly overhead	APAC electronics supplier → China plant	800 km road supplier → CN plant
1.4.1	2	1.4	Driver PCB SHPC-Z JXL 3086	FR4 PCB + SMT parts	0.0024	0.0024	13.50	0.0324	0.0324	FR4 fabrication + SMT assembly	APAC PCB supplier	800 km road to CN assembly
1.4.1.1	3	1.4.1	FR4 laminate share	Glass fiber + epoxy + copper + trace finish meta	0.00110	0.00110	13.18	0.0145	0.0145	Lamination + copper cladding + etching	Integrated upstream	Included upstream
1.4.1.1.1	4	1.4.1.1	Glass fiber fraction	Glass fiber	0.00050	0.00050	10.00	0.0050	0.0050	Fiber production	Materials supplier	Upstream integrated
1.4.1.1.2	4	1.4.1.1	Epoxy fraction	Epoxy resin	0.00030	0.00030	15.00	0.0045	0.0045	Resin synthesis	Chemical supplier	Upstream integrated
1.4.1.1.3	4	1.4.1.1	Copper layer	Copper foil	0.00025	0.00025	12.50	0.0031	0.0031	Copper cladding	Metal supplier	Upstream integrated
1.4.1.1.4	4	1.4.1.1	Nickel finish trace	Nickel	0.00003	0.00003	20.00	0.0006	0.0006	Surface finishing / plating	PCB supplier	Upstream integrated
1.4.1.1.5	4	1.4.1.1	Gold finish trace	Gold	0.00002	0.00002	40.00	0.0008	0.0008	Surface finishing	PCB supplier	Upstream integrated
1.4.1.2	3	1.4.1	Semiconductor / rectifier / control IC share	Silicon + mold compound + trace noble metals	0.00042	0.00042	26.19	0.0110	0.0110	Semiconductor packaging	Integrated upstream	Included upstream
1.4.1.2.1	4	1.4.1.2	Silicon / mold compound fraction	Silicon + mold compound	0.00038	0.00038	25.00	0.0095	0.0095	Semiconductor packaging	Integrated upstream	Included upstream
1.4.1.2.2	4	1.4.1.2	Silver contact trace	Silver	0.00003	0.00003	25.00	0.0008	0.0008	Metallization	Integrated upstream	Included upstream
1.4.1.2.3	4	1.4.1.2	Palladium trace	Palladium	0.00001	0.00001	45.00	0.0005	0.0005	Metallization	Integrated upstream	Included upstream
1.4.1.3	3	1.4.1	Passive component share	Ceramic + metals	0.00050	0.00050	11.40	0.0057	0.0057	Passive component manufacture	Integrated upstream	Included upstream
1.4.1.3.1	4	1.4.1.3	Ceramic body share	Ceramic	0.00030	0.00030	10.00	0.0030	0.0030	Passive component manufacture	Integrated upstream	Included upstream

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1.4.1.3.1	4	1.4.1.3	Ceramic body share	Ceramic	0.00030	0.00030	10.00	0.0030	0.0030	Passive component manufacture	Integrated upstream	Included upstream
1.4.1.3.2	4	1.4.1.3	Base metal share	Nickel / copper	0.00015	0.00015	12.00	0.0018	0.0018	Metallization / forming	Integrated upstream	Included upstream
1.4.1.3.3	4	1.4.1.3	Silver termination trace	Silver	0.00005	0.00005	18.00	0.0009	0.0009	Termination metallization	Integrated upstream	Included upstream
1.4.1.4	3	1.4.1	Solder / flux share	Solder + flux + minor silver	0.00038	0.00038	6.58	0.0025	0.0025	Reflow soldering	Integrated upstream	Included upstream
1.4.1.4.1	4	1.4.1.4	Tin solder fraction	Tin	0.00033	0.00033	6.00	0.0020	0.0020	Solder alloy preparation	Integrated upstream	Included upstream
1.4.1.4.2	4	1.4.1.4	Silver solder fraction	Silver	0.00003	0.00003	25.00	0.0008	0.0008	Solder alloy preparation	Integrated upstream	Included upstream
1.4.1.4.3	4	1.4.1.4	Flux / organics fraction	Flux / organics	0.00002	0.00002	2.50	0.0001	0.0001	Flux formulation	Integrated upstream	Included upstream
1.4.2	2	1.4	Electrolytic capacitor	Aluminium + electrolyte + paper	0.0010	0.0010	18.00	0.0180	0.0180	Capacitor winding + filling + sealing	APAC capacitor supplier	800 km road to CN assembly
1.4.2.1	3	1.4.2	Aluminium can / foil share	Aluminium	0.0005	0.0005	18.00	0.0090	0.0090	Foil rolling + can forming	Capacitor supplier	Upstream integrated
1.4.2.2	3	1.4.2	Electrolyte share	Chemical electrolyte	0.0002	0.0002	18.00	0.0036	0.0036	Chemical formulation	Chemical supplier	Upstream integrated
1.4.2.3	3	1.4.2	Paper separator share	Cellulosic paper	0.0001	0.0001	18.00	0.0018	0.0018	Paper converting	Paper supplier	Upstream integrated
1.4.2.4	3	1.4.2	Rubber / seal share	Elastomer	0.0002	0.0002	18.00	0.0036	0.0036	Elastomer molding	Polymer supplier	Upstream integrated
1.4.3	2	1.4	Driver housing / plastic guide	Plastic "4" observed	0.0014	0.0014	2.30	0.0032	0.0032	Injection molding	APAC plastics supplier	500 km road to CN assembly
1.4.4	2	1.4	Additional driver overhead / assembly burden	Electricity, auxiliaries, scrap, waste handling	0.0000	0.0000	-	0.0289	0.0289	SMT energy + test + process losses	Signify / supply-chain overhead	Modeled as plant / process burden
1.5	1	1.0	E27 cap assembly	Metal + ceramic	0.0000	0.0022	-	0.0000	0.0146	Cap forming + contact assembly	Standard E27 supplier → China plant	500 km road supplier → CN plant
1.5.1	2	1.5	Screw shell E27	Aluminium / thin metal shell	0.0014	0.0014	8.50	0.0119	0.0119	Stamping + forming	APAC cap supplier	500 km road to CN assembly
1.5.2	2	1.5	Bottom contact	Copper alloy	0.0003	0.0003	4.00	0.0012	0.0012	Stamping	APAC cap supplier	500 km road to CN assembly
1.5.3	2	1.5	Electrical insulator	Ceramic / thermoset	0.0005	0.0005	3.00	0.0015	0.0015	Pressing + sintering / molding	APAC ceramic supplier	500 km road to CN assembly
1.6	1	1.0	Internal wiring	Metal	0.0000	0.0007	-	0.0000	0.0028	Wire cutting + soldering	Standard bulb architecture	300 km road supplier → CN plant
1.6.1	2	1.6	Internal wires	Copper / tinned copper	0.0007	0.0007	4.00	0.0028	0.0028	Wire drawing + cutting	APAC wire supplier	300 km road to CN assembly
1.7	1	1.0	Packaging allocation per bulb	Paper + PE	0.0000	0.0008	-	0.0000	0.0010	Carton conversion + film extrusion	Consumer pack allocation per bulb	100 km road supplier → CN plant
1.7.1	2	1.7	Cardboard share per bulb	Paperboard	0.0006	0.0006	0.90	0.0005	0.0005	Die cutting + printing	Packaging supplier	100 km road to CN assembly
1.7.2	2	1.7	Plastic film share per bulb	PE	0.0002	0.0002	2.30	0.0005	0.0005	Film extrusion	Packaging supplier	100 km road to CN assembly

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Item No.	Level	Parent Item	Description	Material	Direct mass (kg)	Roll-up mass (kg)	kgCO2e/kg	kgCO2e/part	Roll-up kgCO2e	Manufacturing method (LCA style)	Supplier model	Shipping detail
1.0	0	-	Philips LED A60 E27 bulb, SKU 8720169191655	Mixed	0.0000	0,0295	-	0.0000	0,1615	Final assembly + electrical test + pack-out	Signify China final assembly	Consumer SKU assembled in China; downstream
1.1	1	1.0	Plastic optical envelope assembly	Polymer	0.0000	0,0120	-	0.0000	0,0397	Optical dome molding + bonding + internal plastic supports	APAC plastics supplier → China plant	500 km road supplier → CN plant
1.1.1	2	1.1	Outer diffuser dome	Polycarbonate / diffuser polymer	0.0085	0,0100	3.50	0.0298	0,0351	Injection molding + surface frosting	APAC plastics supplier	500 km road to CN assembly
1.1.1.1	3	1.1.1	Diffuser polymer feed	Polycarbonate family	0.0000	0,0085	-	0.0000	0,0298	Polymer feed preparation	Integrated upstream	Included upstream
1.1.1.1.1	4	1.1.1.1	Base polymer fraction	PC-like fossil polymer	0.0078	0,0078	3.50	0.0273	0,0273	Polymerization	Polymer supplier	Upstream integrated
1.1.1.1.2	4	1.1.1.1	Diffusion / opacity additives	Mineral filler / additive package	0.0007	0,0007	3.50	0.0025	0,0025	Compounding	Additive supplier	Upstream integrated
1.1.2	2	1.1	Inner optical shell / mixing layer	Diffuser polymer	0.0015	0,0015	3.50	0.0053	0,0053	Thin-wall injection molding	APAC plastics supplier	500 km road to CN assembly
1.1.3	2	1.1	Neck bonding / seal	Epoxy adhesive	0	0	0	0	0	Adhesive dispensing + curing	APAC chemical supplier	500 km road to CN assembly
1.1.4	2	1.1	LED board plastic support ring	Plastic "4" observed	0.0012	0,0012	2.30	0.0028	0,0028	Injection molding	APAC plastics supplier	500 km road to CN assembly
1.1.5	2	1.1	Driver PCB plastic holder / inner chassis	Plastic "4" observed	0.0008	0,0008	2.30	0.0018	0,0018	Injection molding	APAC plastics supplier	500 km road to CN assembly
1.2	1	1.0	Thermal / structural module	Metal + polymer	0.0000	0,0047	-	0.0000	0,0071	Internal heat spreading + structural support	APAC supplier → China plant	500 km road supplier → CN plant
1.2.1	2	1.2	Internal heat spreader / metal cup	Aluminium alloy	0.0035	0,0035	8.50	0.0298	0,0043	Stamping / deep drawing / forming	APAC metal supplier	500 km road to CN assembly
1.2.1.1	3	1.2.1	Aluminium stock	Aluminium sheet / cup stock	0.0000	0,0035	-	0.0000	0,0043	Sheet preparation	Integrated upstream	Included upstream
1.2.1.1.1	4	1.2.1.1	Recycled aluminium share	Recycled aluminium	0.0033	0,0033	0.80	0.0026	0,0026	Secondary aluminium smelting + rolling	Aluminium recycler	Upstream integrated
1.2.1.1.2	4	1.2.1.1	Alloying fraction	Si / Mg alloying	0.0002	0,0002	8.50	0.0017	0,0017	Alloying	Alloy supplier	Upstream integrated
1.2.2	2	1.2	Secondary internal support	Plastic "4" observed	0.0012	0,0012	2.30	0.0028	0,0028	Injection molding	APAC plastics supplier	500 km road to CN assembly
1.3	1	1.0	LED light engine assembly	Electronics + polymer	0.0000	0,00429	-	0.0000	0,03564	LED board + optical interface + locator	APAC LED supplier → China plant	800 km road supplier → CN plant
1.3.1	2	1.3	LED board SHPC-Z JXL 4421	MCPCB + LEDs	0.0022	0,00219	15.09	0.0332	0,02564	MCPCB fabrication + SMT LED mounting	APAC LED board supplier	800 km road to CN assembly
1.3.1.1	3	1.3.1	MCPCB substrate share	Aluminium + dielectric + copper + trace finish n	0.0012	0,0012	10.67	0.0128	0,00594	Aluminium-core PCB manufacture	Integrated upstream	Included upstream
1.3.1.1.1	4	1.3.1.1	Aluminium core (recycled)	Recycled aluminium	0.00078	0,00078	0.80	0.0006	0,0006	Sheet rolling / punching	Metal supplier	Upstream integrated
1.3.1.1.2	4	1.3.1.1	Bio-based dielectric layer	Thermally conductive bio-resin	0.00010	0,0001	6.00	0.0006	0,0006	Coating / lamination	PCB supplier	Upstream integrated
1.3.1.1.3	4	1.3.1.1	Copper circuit layer	Copper foil	0.00029	0,00029	14.00	0.0041	0,0041	Copper lamination + etching	PCB supplier	Upstream integrated
1.3.1.1.4	4	1.3.1.1	Nickel finish trace	Nickel	0.00002	0,00002	20.00	0.0004	0,0004	Surface finishing / plating	PCB supplier	Upstream integrated
1.3.1.1.5	4	1.3.1.1	Reduced gold finish trace	Gold	0.000002	0,000002	40.00	0.00008	0,00008	Surface finishing	PCB supplier	Upstream integrated
1.3.1.1.6	4	1.3.1.1	Copper finish share	Copper	0.000004	0,000004	14.00	0.00006	0,00006	Surface finishing	PCB supplier	Upstream integrated
1.3.1.1.7	4	1.3.1.1	Silver finish share	Silver	0.000004	0,000004	25.00	0.00010	0,0001	Surface finishing	PCB supplier	Upstream integrated
1.3.1.2	3	1.3.1	LED packages share	Semiconductor + package metals + silicone/re	0.00070	0,00069	25.86	0.0181	0,0175	LED package assembly	Integrated upstream	Included upstream
1.3.1.2.1	4	1.3.1.2	Semiconductor die share	Gallium high grade / semiconductor proxy	0.00020	0,0002	35.00	0.0070	0,007	Epitaxy + dicing	Semiconductor supplier	Upstream integrated
1.3.1.2.2	4	1.3.1.2	Leadframe / package metals	Copper	0.00018	0,00018	20.00	0.0036	0,0036	Stamping + plating	Package supplier	Upstream integrated
1.3.1.2.3	4	1.3.1.2	Package resin / lens	Silicone / resin	0.00028	0,00028	21.67	0.0061	0,0061	Dispensing + curing	Package supplier	Upstream integrated
1.3.1.2.4	4	1.3.1.2	Silver interconnect trace	Silver	0.00003	0,00003	25.00	0.0008	0,0008	Wire bonding / metallization	Package supplier	Upstream integrated
1.3.1.3	3	1.3.1	Solder / interface share	Sn-based solder + minor silver	0.00030	0,0003	7.33	0.0022	0,0022	Reflow soldering	Integrated upstream	Included upstream
1.3.1.3.1	4	1.3.1.3	Tin solder fraction	Tin	0.00028	0,00028	6.00	0.0017	0,0017	Solder alloy preparation	Integrated upstream	Included upstream
1.3.1.3.2	4	1.3.1.3	Silver solder fraction	Silver	0.00002	0,00002	25.00	0.0005	0,0005	Solder alloy preparation	Integrated upstream	Included upstream
1.3.2	2	1.3	Optical silicone / phosphor interface	Silicone + yttrium oxide	0.0010	0,001	7.50	0.0075	0,0075	Coating / dispensing + cure	APAC LED supplier	800 km road to CN assembly
1.3.2.1	3	1.3.2	Optical silicone fraction	Silicone	0.00085	0,00085	6.50	0.0055	0,0055	Dispensing + cure	APAC LED supplier	Included upstream
1.3.2.2	3	1.3.2	Phosphor fraction	Yttrium oxide	0.00015	0,00015	13.33	0.0020	0,002	Phosphor mixing / coating	Chemical supplier	Included upstream
1.3.3	2	1.3	LED board plastic locator	Plastic "4" observed	0.0011	0,0011	2.30	0.0025	0,0025	Injection molding	APAC plastics supplier	500 km road to CN assembly

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1.4	1 1.0	Driver electronics assembly	Electronics	0.0000	0,00479	-	0.0000	0,07147	Driver PCB + capacitor + plastic guide + assembly overhead	APAC electronics supplier → China plant	800 km road supplier → CN plant
1.4.1	2 1.4	Driver PCB SHPC-Z JXL 3086	FR4 PCB + SMT parts	0.0024	0,00239	13.50	0.0324	0,02957	FR4 fabrication + SMT assembly	APAC PCB supplier	800 km road to CN assembly
1.4.1.1	3 1.4.1	FR4 laminate share	Glass fiber + epoxy + copper + trace finish meta	0.00110	0,0011	13.18	0.0145	0,01067	Lamination + copper cladding + etching	Integrated upstream	Included upstream
1.4.1.1.1	4 1.4.1.1	Glass fiber fraction	Glass fiber	0.00050	0,0005	10.00	0.0050	0,005	Fiber production	Materials supplier	Upstream integrated
1.4.1.1.2	4 1.4.1.1	Biodegradable resin fraction	Bio-based epoxy / biodegradable resin	0.00030	0,0003	5.00	0.0015	0,0015	Bio-resin synthesis	Chemical supplier	Upstream integrated
1.4.1.1.3	4 1.4.1.1	Copper layer	Copper foil	0.00025	0,00025	12.50	0.0031	0,0031	Copper cladding	Metal supplier	Upstream integrated
1.4.1.1.4	4 1.4.1.1	Nickel finish trace	Nickel	0.00003	0,00003	20.00	0.0006	0,0006	Surface finishing / plating	PCB supplier	Upstream integrated
1.4.1.1.5	4 1.4.1.1	Reduced gold finish trace	Gold	0.000004	0,000004	40.00	0.00016	0,00016	Surface finishing	PCB supplier	Upstream integrated
1.4.1.1.6	4 1.4.1.1	Copper finish share	Copper	0.000008	0,000008	14.00	0.00011	0,00011	Surface finishing	PCB supplier	Upstream integrated
1.4.1.1.7	4 1.4.1.1	Silver finish share	Silver	0.000008	0,000008	25.00	0.00020	0,0002	Surface finishing	PCB supplier	Upstream integrated
1.4.1.2	3 1.4.1	Semiconductor / rectifier / control IC share	Silicon + mold compound + trace noble metals	0.00042	0,00041	26.19	0.0110	0,0103	Semiconductor packaging	Integrated upstream	Included upstream
1.4.1.2.1	4 1.4.1.2	Silicon / mold compound fraction	Silicon + mold compound	0.00038	0,00038	25.00	0.0095	0,0095	Semiconductor packaging	Integrated upstream	Included upstream
1.4.1.2.2	4 1.4.1.2	Silver contact trace	Silver	0.00003	0,00003	25.00	0.0008	0,0008	Metallization	Integrated upstream	Included upstream
1.4.1.3	3 1.4.1	Passive component share	Ceramic + metals	0.00050	0,0005	11.40	0.0057	0,0057	Passive component manufacture	Integrated upstream	Included upstream
1.4.1.3.1	4 1.4.1.3	Ceramic body share	Ceramic	0.00030	0,0003	10.00	0.0030	0,003	Passive component manufacture	Integrated upstream	Included upstream
1.4.1.3.2	4 1.4.1.3	Base metal share	Nickel / copper	0.00015	0,00015	12.00	0.0018	0,0018	Metallization / forming	Integrated upstream	Included upstream
1.4.1.3.3	4 1.4.1.3	Silver termination trace	Silver	0.00005	0,00005	18.00	0.0009	0,0009	Termination metallization	Integrated upstream	Included upstream
1.4.1.4	3 1.4.1	Solder / flux share	Solder + flux + minor silver	0.00038	0,00038	6.58	0.0025	0,0029	Reflow soldering	Integrated upstream	Included upstream
1.4.1.4.1	4 1.4.1.4	Tin solder fraction	Tin	0.00033	0,00033	6.00	0.0020	0,002	Solder alloy preparation	Integrated upstream	Included upstream
1.4.1.4.2	4 1.4.1.4	Silver solder fraction	Silver	0.00003	0,00003	25.00	0.0008	0,0008	Solder alloy preparation	Integrated upstream	Included upstream
1.4.1.4.3	4 1.4.1.4	Flux / organics fraction	Flux / organics	0.00002	0,00002	2.50	0.0001	0,0001	Flux formulation	Integrated upstream	Included upstream
1.4.2	2 1.4	Electrolytic capacitor	Aluminium + electrolyte + paper	0.0010	0,001	18.00	0.0180	0,0098	Capacitor winding + filling + sealing	APAC capacitor supplier	800 km road to CN assembly
1.4.2.1	3 1.4.2	Aluminium can / foil share (recycled)	Recycled aluminium	0.0005	0,0005	1.50	0.0008	0,0008	Foil rolling + can forming	Capacitor supplier	Upstream integrated
1.4.2.2	3 1.4.2	Electrolyte share	Chemical electrolyte	0.0002	0,0002	18.00	0.0036	0,0036	Chemical formulation	Chemical supplier	Upstream integrated
1.4.2.3	3 1.4.2	Paper separator share	Cellulosic paper	0.0001	0,0001	18.00	0.0018	0,0018	Paper converting	Paper supplier	Upstream integrated
1.4.2.4	3 1.4.2	Rubber / seal share	Elastomer	0.0002	0,0002	18.00	0.0036	0,0036	Elastomer molding	Polymer supplier	Upstream integrated
1.4.3	2 1.4	Driver housing / plastic guide	Plastic "4" observed	0.0014	0,0014	2.30	0.0032	0,0032	Injection molding	APAC plastics supplier	500 km road to CN assembly
1.4.4	2 1.4	Additional driver overhead / assembly burden	Electricity, auxiliaries, scrap, waste handling	0.0000	0	-	0.0289	0,0289	SMT energy + test + process losses	Signify / supply-chain overhead	Modeled as plant / process burden
1.5	1 1.0	E27 cap assembly	Metal + ceramic	0.0000	0,0022	-	0.0000	0,0038	Cap forming + contact assembly	Standard E27 supplier → China plant	500 km road supplier → CN plant
1.5.1	2 1.5	Screw shell E27 (recycled aluminium)	Recycled aluminium	0.0014	0,0014	0.80	0.0011	0,0011	Stamping + forming	APAC cap supplier	500 km road to CN assembly
1.5.2	2 1.5	Bottom contact	Copper alloy	0.0003	0,0003	4.00	0.0012	0,0012	Stamping	APAC cap supplier	500 km road to CN assembly
1.5.3	2 1.5	Electrical insulator	Ceramic / thermoset	0.0005	0,0005	3.00	0.0015	0,0015	Pressing + sintering / molding	APAC ceramic supplier	500 km road to CN assembly
1.6	1 1.0	Internal wiring	Metal	0.0000	0,0007	-	0.0000	0,0028	Wire cutting + soldering	Standard bulb architecture	300 km road supplier → CN plant
1.6.1	2 1.6	Internal wires	Copper / tinned copper	0.0007	0,0007	4.00	0.0028	0,0028	Wire drawing + cutting	APAC wire supplier	300 km road to CN assembly
1.7	1 1.0	Packaging allocation per bulb	Paper + PE	0.0000	0,0008	-	0.0000	0,001	Carton conversion + film extrusion	Consumer pack allocation per bulb	100 km road supplier → CN plant
1.7.1	2 1.7	Cardboard share per bulb	Paperboard	0.0006	0,0006	0.90	0.0005	0,0005	Die cutting + printing	Packaging supplier	100 km road to CN assembly
1.7.2	2 1.7	Plastic film share per bulb	PE	0.0002	0,0002	2.30	0.0005	0,0005	Film extrusion	Packaging supplier	100 km road to CN assembly

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Life cycle Assessment

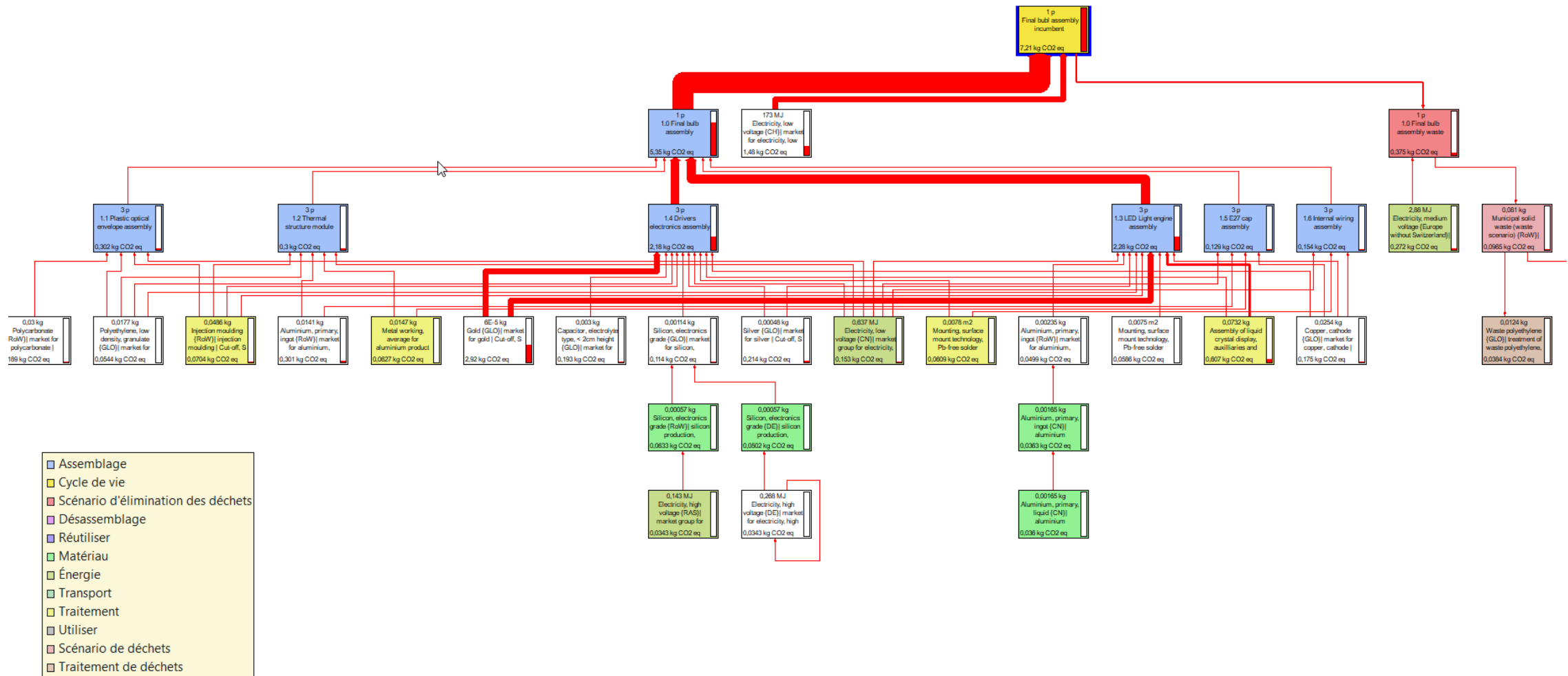
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Mathélyo Favre-Régouillon

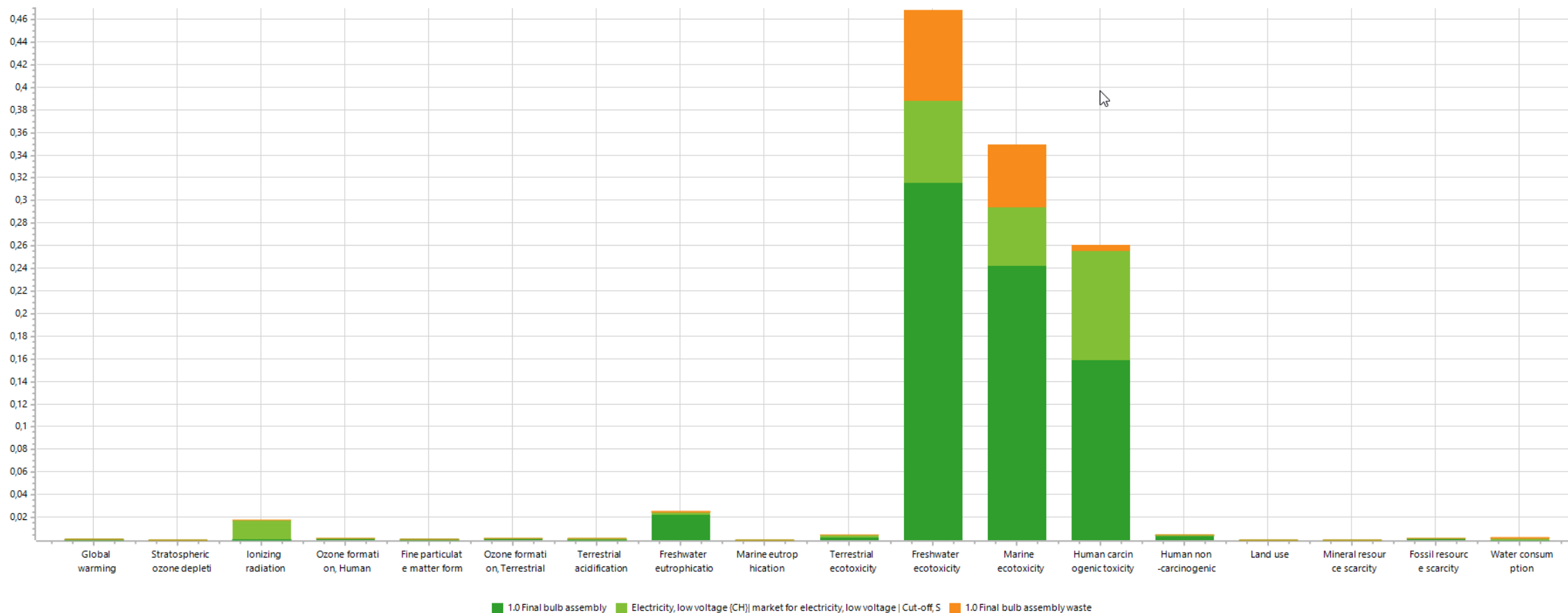
Philips LED E27 Bulb

1. Cycle E-Sankey diagram --- Midpoint (H)



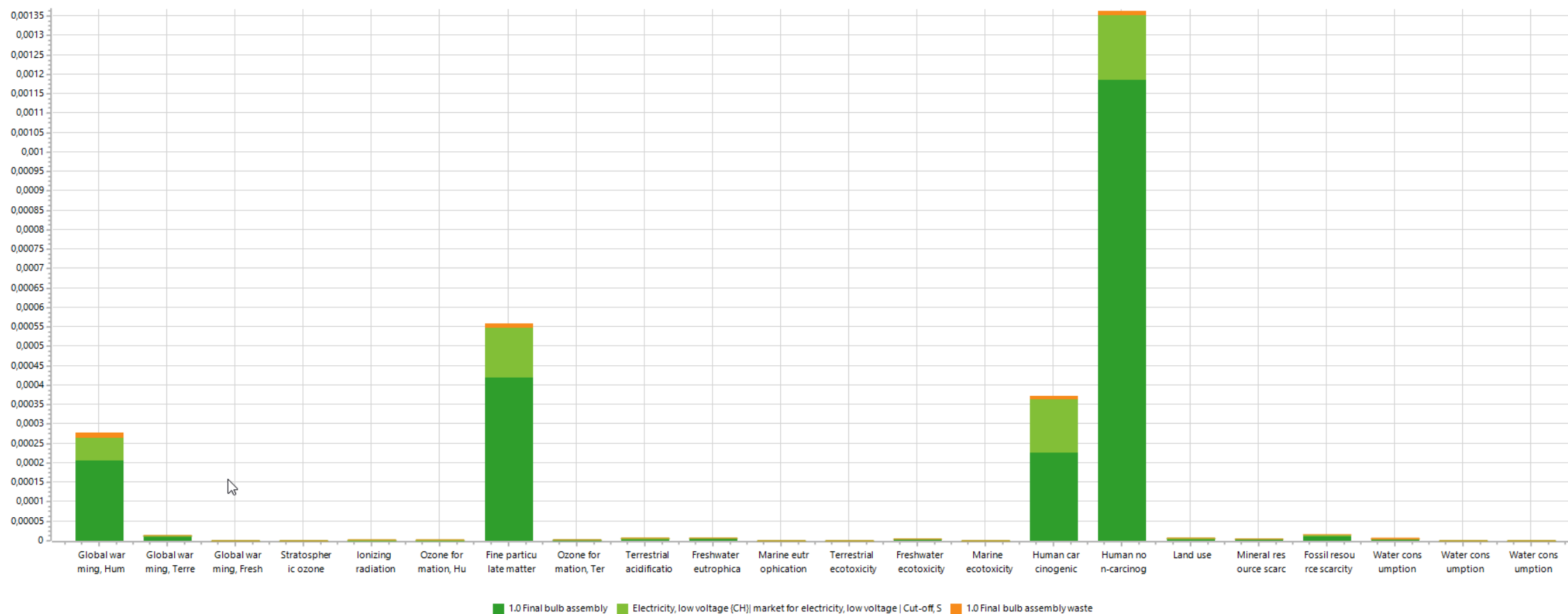
Philips LED E27 Bulb

1. Impact study --- Midpoint (H) – Full life cycle



Philips LED E27 Bulb

1. Impact study --- Endpoint (H) – Full life cycle



Philips LED E27 Bulb

1. Impact study --- Endpoint (H) – Full life cycle



Philips LED E27 Bulb

1. Impact study --- Midpoint (H) – Only Production phase

